

Answers

1. Compute $(1, 0, 0) \times (0, 1, 0)$

Answer: $(0, 0, 1)$

2. Compute $(2, 2, 2) \times (0, 1, 0)$

Answer: $(-2, 0, 2)$

3. Compute $(1, 2, 3) \times (4, 5, 6)$

Answer: $(-3, 6, -3)$

4. Given $\hat{a}$ and $\hat{b}$, explain the geometric meaning of $\hat{a} \times \hat{b}$.

Answer: $C = \hat{a} \times \hat{b}$, C is a vector perpendicular to the plane formed by $\hat{a}$ and $\hat{b}$.
Additionally, C is orthogonal to both $\hat{a}$ and $\hat{b}$.

5. Given $\vec{a}$ and $\vec{b}$, what does the magnitude of $\vec{a} \times \vec{b}$ represent?

Answer: The magnitude of $\vec{a} \times \vec{b}$ represents the area of the parallelogram spanned by the vectors.

6. Given $C = \hat{a} \times \hat{b}$, what happens if you switch the order of the arguments, ie. $\hat{b} \times \hat{a}$?

Answer: If $C = \hat{b} \times \hat{a}$, then if you switch the order of the arguments, the resulting vector would be $-C$, it would go in the opposite direction.

7. If $\hat{a}$ and $\hat{b}$ are parallel, what is $\hat{a} \times \hat{b}$?

Answer: If they're parallel, the result is the zero vector.

8. If $\vec{a} \times \vec{b} = 0$, what does this mean?

Answer: Either they're parallel, or at least one of them is the zero vector.

9. What is the result of $\vec{a} \times \vec{a}$?

Answer: The vectors are parallel, and thus the result is the zero vector.

10. Suppose you have 3 points in space, P_1, P_2, P_3 . How could you use the cross product to find a normal to the plane that passes through these 3 points?

Answer: You could calculate 2 vectors from the 3 points, so $V_1 = P_2 - P_1$, and $V_2 = P_3 - P_1$, and then calculate $V_1 \times V_2$. The result would be normal to the plane that passes through those 3 points.

11. How could you use the cross product to calculate the surface area of a triangle with 3 vertices, A, B, and C?

Answer: This is extremely similar to the previous question, but uses a different property of the cross product. To begin, you'd compute 2 vectors from the 3 points, let's say you calculate $AB = B - A$, and $AC = C - A$. Then $N = AB \times AC$, and then $\frac{1}{2} |N|$ would be the area of the triangle, because the magnitude of N is equal to the area of the parallelogram spanned by AB and AC .